



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COMPILER DESIGN (24CS2235F)

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TITLE OF THE PROJECT	Syntax Error Detection and Recovery System	
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Abstract

The **Syntax Error Detection and Recovery System** is a compiler-based application designed to detect, analyze, and recover from syntax errors in source code. The system performs lexical analysis and syntax analysis to convert source code into tokens and verify whether the program follows a predefined grammar. It identifies common syntax errors such as missing semicolons, unmatched brackets, missing parentheses, unexpected tokens, and invalid statement structures. For every detected error, the system provides the error location, type, expected token, and a meaningful explanation to help the user understand the problem.

A key feature of the system is its **error recovery mechanism**, which allows the compiler to continue analyzing the source program instead of terminating at the first syntax error. Classical compiler techniques such as **panic-mode recovery**, **phrase-level recovery**, and **error productions** are used to skip or correct invalid portions of the input and resume parsing from an appropriate point. The system can also generate a parse tree and recovered source representation, helping users visualize how the parser processes and corrects erroneous input.

The proposed system provides an interactive and educational platform for understanding syntax analysis and error handling in compiler design. It helps users identify multiple syntax errors in a single program, understand the reason behind each error, and observe the recovery process. The project demonstrates practical implementation of important compiler concepts while providing scope for future enhancements such as automated correction suggestions, advanced parsing techniques, support for multiple programming languages, and graphical visualization of compiler phases.

Signature of the Guide

HOD